

ADC SerDes Tutorial

Feed-forward equalization, decision-feedback equalization, and maximum-likelihood sequence estimation

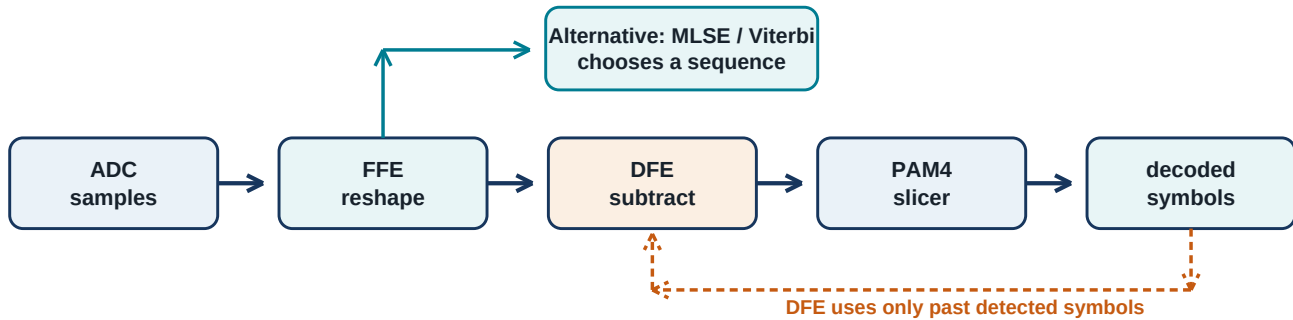


Fig. 1 - Two common digital receiver choices after an ADC.

Purpose

This tutorial builds intuition for the three main digital tools used after an ADC in a high-speed wireline receiver. It uses PAM4 examples, but the ideas apply equally to NRZ and other multi-level formats.

Key idea: an FFE reshapes the channel, a DFE subtracts echoes from already detected symbols, and MLSE makes a delayed decision by comparing complete candidate sequences.

Learning goals

- Read a symbol-spaced pulse response and identify main cursor, precursor ISI, and postcursor ISI.
- Understand why FFE can cancel precursor ISI but may enhance noise.
- Understand why DFE is efficient for postcursor ISI but can create error bursts.
- Understand when a short-memory MLSE detector can outperform aggressive equalization.

Notation

$a[n]$ is the transmitted PAM4 symbol in $\{-3, -1, +1, +3\}$; $r[n]$ is the ADC sample; $h[k]$ is the channel cursor; $w[n]$ is additive noise; and $\hat{a}[n]$ is a receiver decision.

1. The common channel model

At the ADC sampling times, a linear channel is represented by a weighted sum of symbols. The current symbol is not observed in isolation because energy from nearby symbols overlaps it.

$$r[n] = \sum(k=-K_{pre} \text{ to } K_{post}) h[k] a[n-k] + w[n] \quad (\text{Eq. 1})$$



Fig. 2 - Symbol-spaced channel pulse. Energy before 0 UI is precursor ISI; later energy is postcursor ISI.

How to read Fig. 2

- The main cursor $h[0]$ carries the desired symbol $a[n]$.
- Precursor cursors arrive before the main pulse. They depend on future-relative symbols when the decision is referenced to $a[n]$.
- Postcursor cursors are trailing echoes from earlier symbols. They depend on $a[n-1]$, $a[n-2]$, and so on.
- Noise cannot be removed deterministically. A good equalizer trades residual ISI against noise enhancement.

Worked PAM4 sample

Use the channel $h[-1]=0.20$, $h[0]=1.00$, $h[1]=0.45$, $h[2]=-0.20$. For $a[n+1]=-3$, $a[n]=+1$, $a[n-1]=+3$, and $a[n-2]=-1$:

$$r[n] = 0.20(-3) + 1.00(+1) + 0.45(+3) - 0.20(-1) = 1.95 \quad (\text{Eq. 2})$$

The wanted contribution is only +1.00. The remaining +0.95 is structured ISI. The receiver methods below differ in how they deal with that structured interference.

2. Feed-forward equalization (FFE)

An FFE is an FIR filter applied directly to ADC samples. It does not depend on prior hard decisions, so it cannot propagate a wrong slicer decision through feedback.

$$y[n] = \sum_{m=0}^{NF-1} c[m] r[n-m] \quad (\text{Eq. 3})$$

Decision delay makes future-relative samples available causally.

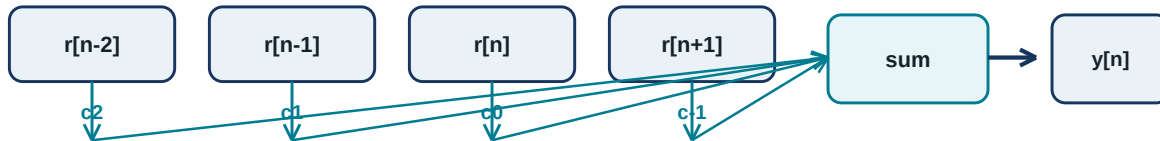


Fig. 3 - FFE is a weighted sum of ADC samples. It can reshape precursor and postcursor ISI, but may enhance noise.

Why FFE can cancel precursor ISI

The receiver delays the decision for $a[n]$. By the time it makes that delayed decision, samples that were future-relative to $a[n]$ have arrived. This is why a symmetric analysis FFE can be implemented causally with a matching output latency.

Zero forcing versus MMSE

A zero-forcing FFE tries to make the combined response $c \cdot h$ look like one isolated cursor. An MMSE FFE accepts some residual ISI when removing it would excessively amplify noise.

$$c_{\text{MMSE}} = \arg \min_c E\{|a[n-d] - c^T r[n]|^2\} \quad (\text{Eq. 4})$$

$$\text{For white input noise: } \sigma_{\text{out}}^2 = \sigma_w^2 \sum(m) |c[m]|^2 \quad (\text{Eq. 5})$$

Eq. 5 is the FFE penalty: large coefficients can improve the pulse response while degrading actual BER. For long, high-loss channels, MMSE is normally the useful baseline.

Practical FFE notes

- Use known training data, least squares, LMS, or NLMS to obtain coefficients.
- T-spaced taps are simple; T/2-spaced taps are more tolerant of timing phase and fractional-delay effects.
- Include coefficient limits, rounding, saturation and output latency before using FFE results for hardware optimization.

3. Decision-feedback equalization (DFE)

A DFE uses already detected symbols to predict and subtract the trailing channel echoes. It is especially efficient when postcursor ISI is large.

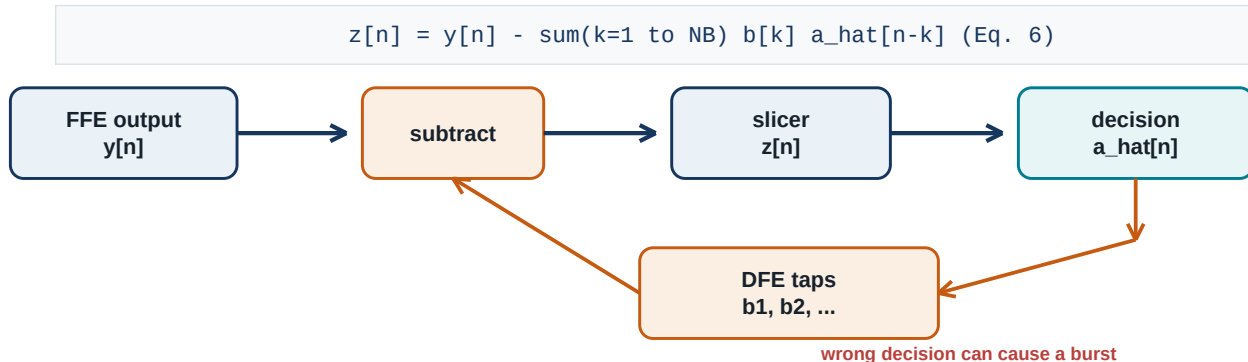


Fig. 4 - DFE removes predicted postcursor ISI using past hard decisions. This avoids FFE noise enhancement but introduces feedback risk.

What a DFE can and cannot cancel

- If the effective postcursor is $g[1]=0.45$, a first DFE tap near $+0.45$ removes $0.45 a_{\text{hat}}[n-1]$.
- The DFE avoids the linear noise enhancement in Eq. 5 because it subtracts reconstructed symbol interference.
- A conventional DFE cannot cancel precursor ISI, because future symbols have not yet been detected.

Error propagation example

Suppose $b[1]=0.45$ and the correct past symbol is $+3$, but the slicer decides $+1$. The correct feedback is $0.45(3)=1.35$; the applied feedback is $0.45(1)=0.45$. The next decision has a residual error of $+0.90$.

$$\text{feedback error} = b[1] (a[n-1] - a_{\text{hat}}[n-1]) = 0.45(3 - 1) = 0.90 \quad (\text{Eq. 7})$$

One wrong decision can therefore increase the chance of later wrong decisions. This creates burst errors, which matter directly to RS-FEC codeword failure probability.

Training versus payload operation

During training, fit DFE taps with known transmitted symbols. During payload operation, replace those symbols with a_{hat} . Always report both genie-aided and decision-directed results; only the decision-directed result is a credible receiver result.

4. Maximum-likelihood sequence estimation (MLSE)

MLSE does not try to eliminate every residual cursor. Instead, it asks which sequence of PAM4 symbols best explains the received samples given a short channel model.

$$y[n] = \sum_{k=0 \text{ to } L} g[k] a[n-k] + v[n] \quad (\text{Eq. 8})$$

$$\hat{a} = \arg \min_a \sum(n) |y[n] - \sum_{k=0 \text{ to } L} g[k] a[n-k]|^2 / \sigma_v^2 \quad (\text{Eq. 9})$$

One-memory PAM4 MLSE has four states: the previous PAM4 symbol.

survivor path: lowest accumulated prediction error

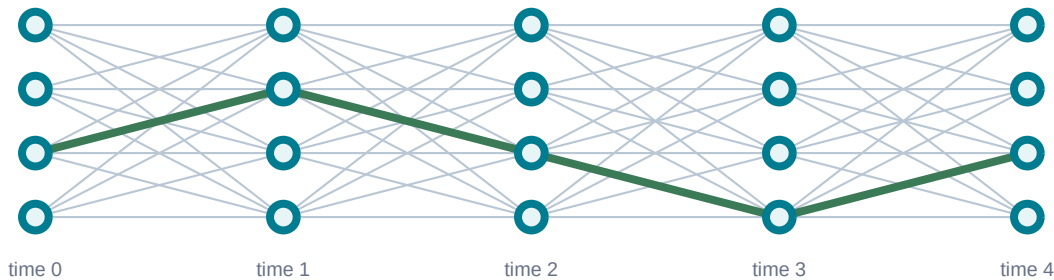


Fig. 5 - Simplified Viterbi trellis. MLSE retains the best candidate path entering each state instead of committing immediately.

The Viterbi idea

- A state stores the last L candidate PAM4 symbols.
- For each possible new PAM4 symbol, predict the next received sample and calculate its squared-error branch metric.
- Accumulate metrics and keep the lowest-metric survivor entering each state.
- Trace back through surviving paths after a delay to output reliable decisions.

State count

Residual memory L	PAM4 states	Branches / detected symbol	Typical use
1	4	16	Strong single postcursor
2	16	64	Two deliberate residual cursors
3	64	256	Only if gain justifies cost
4	256	1,024	Usually expensive for a per-lane receiver

MLSE complexity grows exponentially with memory. This is why the FFE before MLSE should shorten the long physical channel into a deliberately short target response.

5. Channel shortening and receiver choices

A slicer-oriented FFE tries to produce one isolated cursor. An MLSE-oriented FFE instead aims for a short controlled response, such as $[1, 0.35]$. The detector then uses that known remaining memory.

$c*h$ approximately equals $[0, \dots, 0, g[0], g[1], \dots, g[L], 0, \dots]$ (Eq. 10)

Example

Rather than using extreme FFE gain to force $g[1]$ to zero, retain $g=[1, 0.35]$ and use a 4-state, one-memory PAM4 MLSE. This can reduce noise enhancement while preserving detection reliability.

Property	FFE	DFE	MLSE
Primary action	Reshape samples	Subtract past-symbol echoes	Choose most likely path
Precursor ISI	Can cancel with latency	Cannot cancel	Handled with decision delay
Postcursor ISI	Can reduce	Efficiently cancels	Uses known memory
Noise enhancement	Possible	No linear enhancement	Only through any front-end filter
Error propagation	No	Yes	No direct feedback propagation
Complexity	Linear in taps	Linear in feedback taps	Exponential in memory
Best fit	Precursor removal and shaping	Strong early postcursors	Short strong residual memory

Important ADC SerDes insight

The ADC samples are already quantized. Neither FFE, DFE nor MLSE can recover amplitude information lost to clipping, too little ENOB, or a poor recovered clock phase. Equalizer, ADC range, threshold adaptation and CDR must be treated as a coupled system.

Metric selection

- Use dpSNR to understand residual linear error and noise, especially for an FFE.
- Use empirical DER and BER for DFE because error propagation can make residuals non-Gaussian.
- Use BER, error-event length, RS-symbol errors per codeword, and latency/operations for MLSE tradeoffs. An eye alone can underestimate MLSE capability.

6. A practical experiment plan

Use the exact same channel, CTLE, ADC codes, noise realization and held-out data set to compare receiver choices. This prevents a favorable random sequence from looking like an algorithm improvement.

Case	Receiver path	What it teaches
A	ADC -> FFE -> slicer	Baseline residual ISI, noise enhancement, precursor cancellation
B	ADC -> FFE -> DFE -> slicer	Postcursor cancellation and decision-error propagation
C	ADC -> channel-shortening FFE -> 1-memory MLSE	Value of keeping one controlled residual cursor
D	ADC -> channel-shortening FFE -> 2-memory MLSE	BER gain versus 16-state detector cost

Recommended reporting

- FFE coefficients, FFE output noise variance, cursor response and effective group delay.
- DFE genie-aided and decision-directed BER separately, plus error-burst length distribution.
- MLSE target response, state count, branches per UI, traceback depth and BER.
- For RS(544,514), record erroneous 10-bit symbols per codeword, not only raw bit BER.

Recommended first architecture for this model

Keep the current 17-tap code-domain FFE and 12-tap DFE as the conventional baseline. Add a separate channel-shortening FFE mode and compare it with a 4-state one-memory PAM4 MLSE first. Only add a 16-state two-memory MLSE if the first experiment leaves a strong second residual cursor.

Beginner checklist

- FFE first: remove precursor ISI and shape the channel without excessive noise gain.
- DFE next: remove postcursor ISI only when the feedback path is implementable.
- MLSE last: use it for intentionally retained, short channel memory - not for an arbitrarily long tail.
- Do not compare algorithms using only a noiseless eye diagram.
- For ADC-based links, include sampling phase and converter clipping before claiming DSP gain.

End of tutorial